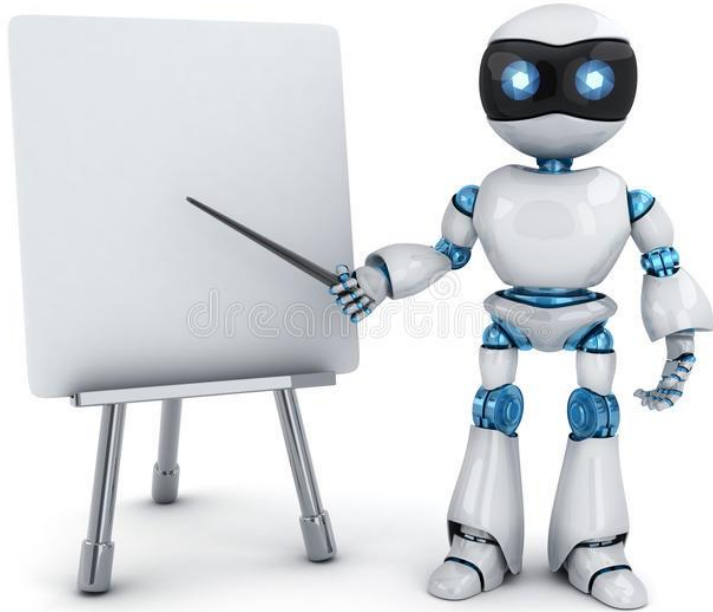




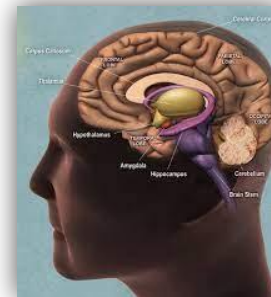
ANN

ARTIFICIAL NEURAL NETWORK

- Artificial neural network algorithm comes under supervised learning and used to solve classification problems.
- ANN algorithm is based on human brain function and used to model complicated pattern and forecast issues.
- The concept of Ann is a deep learning method that arose from the concept of biological neural networks.

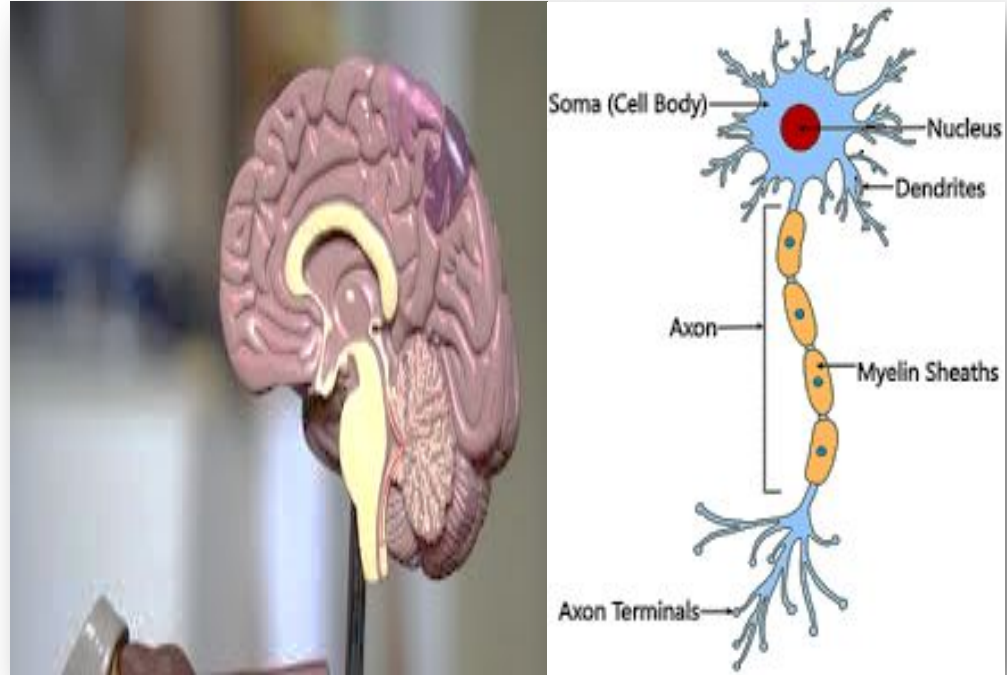
ARTIFICIAL NEURAL NETWORK

- Researchers thought of building a model which mimics human brain.
- The development of ANN was the attempt to replicate the workings of the human brain.
- An artificial neuron network (neural network) is a computational model that mimics the way nerve cells work in the human brain.



ARTIFICIAL NEURAL NETWORK

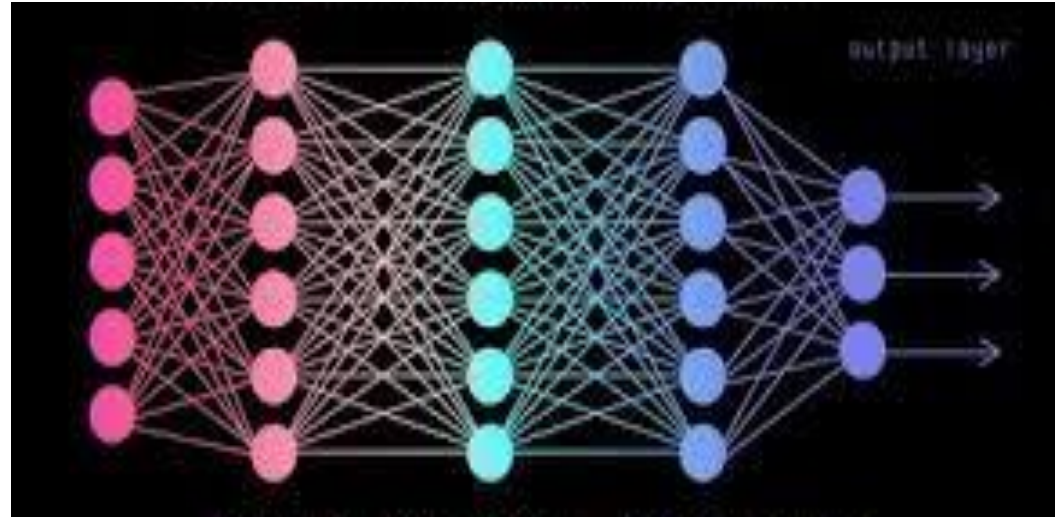
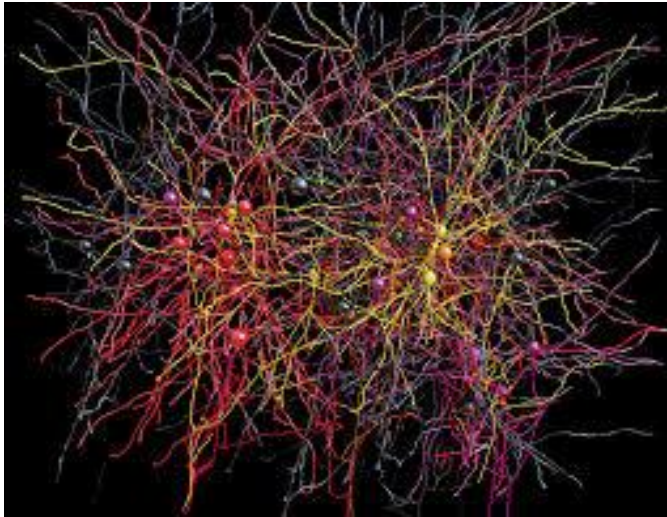
Your brain contains billions of nerve cells arranged in patterns that coordinate thought, emotion, behavior, movement and sensation. A complicated highway system of nerves connects your brain to the rest of your body, so communication can occur in split seconds. Think about how fast you pull your hand back from a hot stove. While all the parts of your brain work together, each part is responsible for a specific function — controlling everything from your heart rate to your mood.



ARTIFICIAL NEURAL NETWORK

- Artificial neural networks are a special type of machine learning algorithms that are modeled after the brain. That is, just like how the neurons in our nervous system are able to learn from the past data. Similarly , ANN is able to learn from data and provide responses in the form of predictions or classifications.
- ANN is non linear model which find complex relationship between the inputs and outputs.
- Mathematical neuron created by human is called **Artificial Neuron**.

ANN ARCHITECHTURE



HOW ANN WORKS?

DIFFERENT LAYERS IN ANN

- Input layer
- Hidden layer
- Output layer

LAYERS

INPUT LAYER

Input layer is the first layer of the neural network that receives the input information such as features in the data.

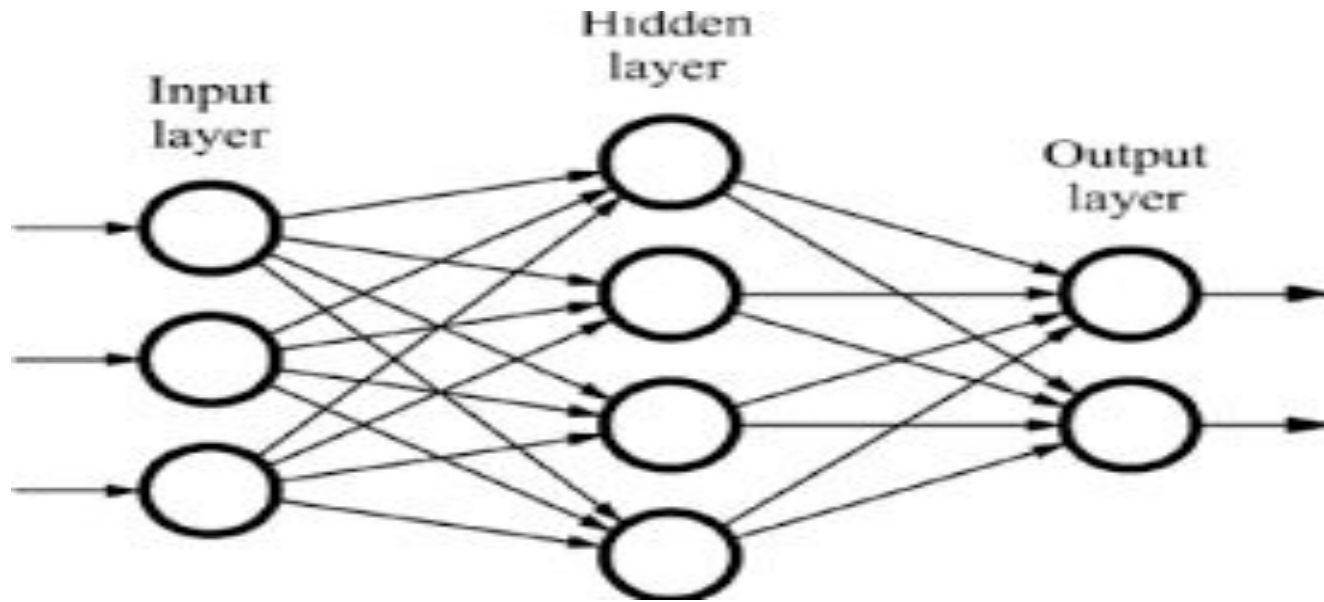
HIDDEN LAYER

Hidden layer is the middle layer where the actual training or learning happens. That means various mathematical computation of input data and pattern recognition happens inside this layer.

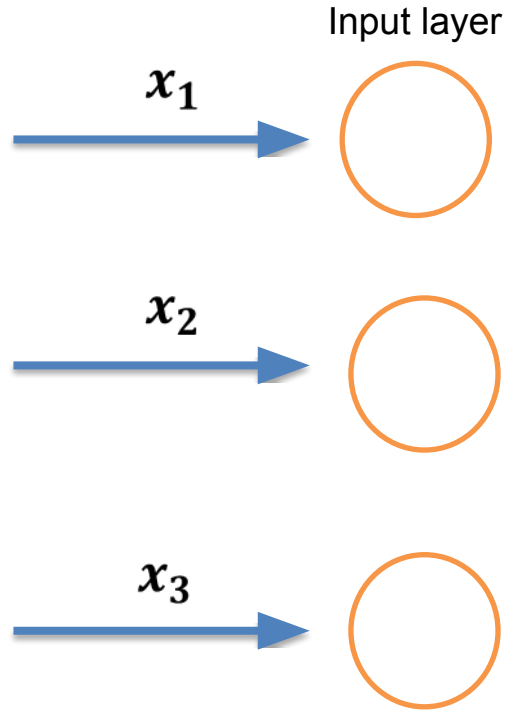
OUTPUT LAYER

This layer gives us the output obtained through rigorous computation performed by the hidden layer.

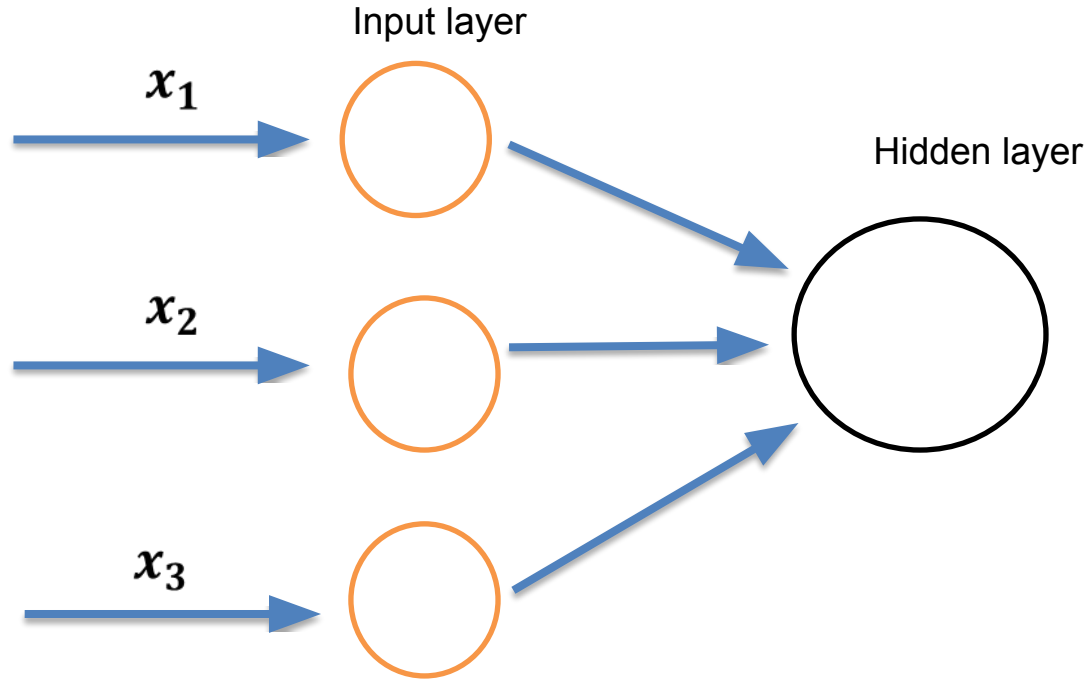
ANN



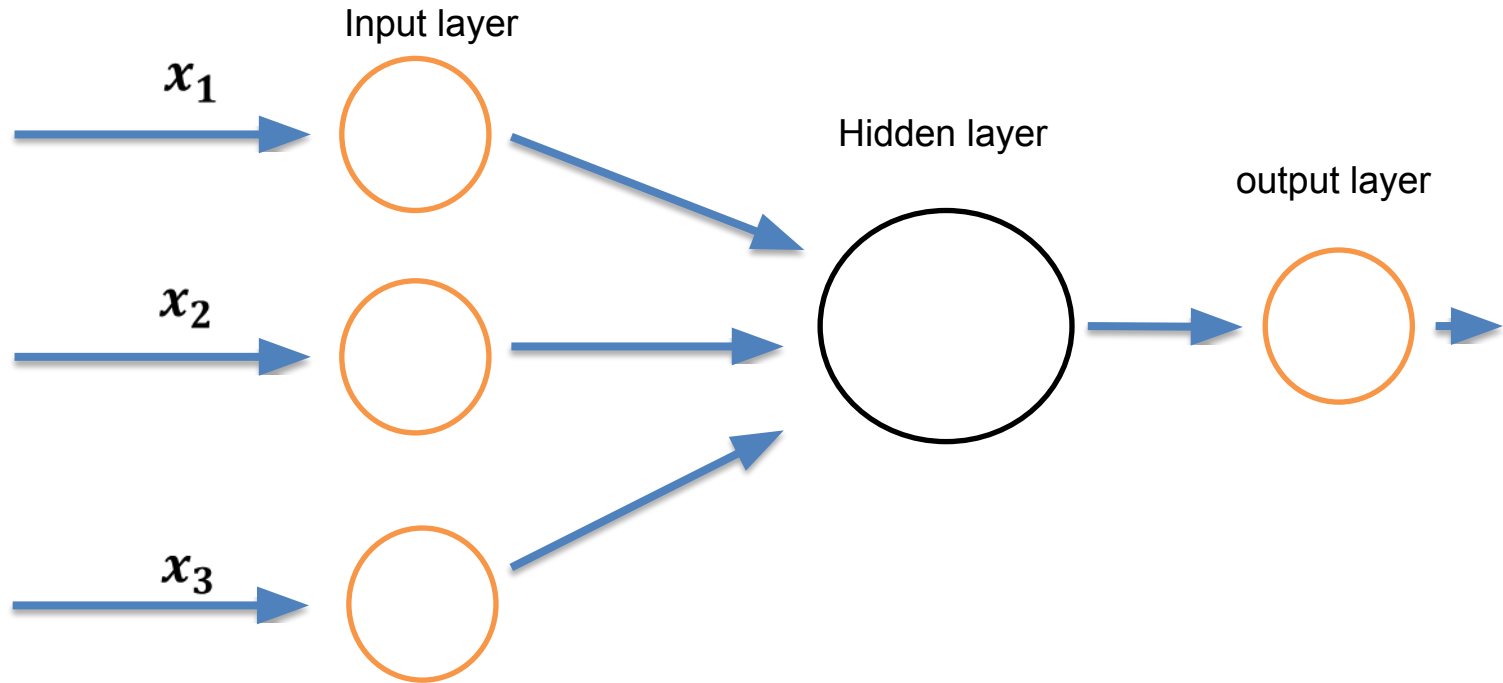
INPUT LAYER



HIDDEN LAYER

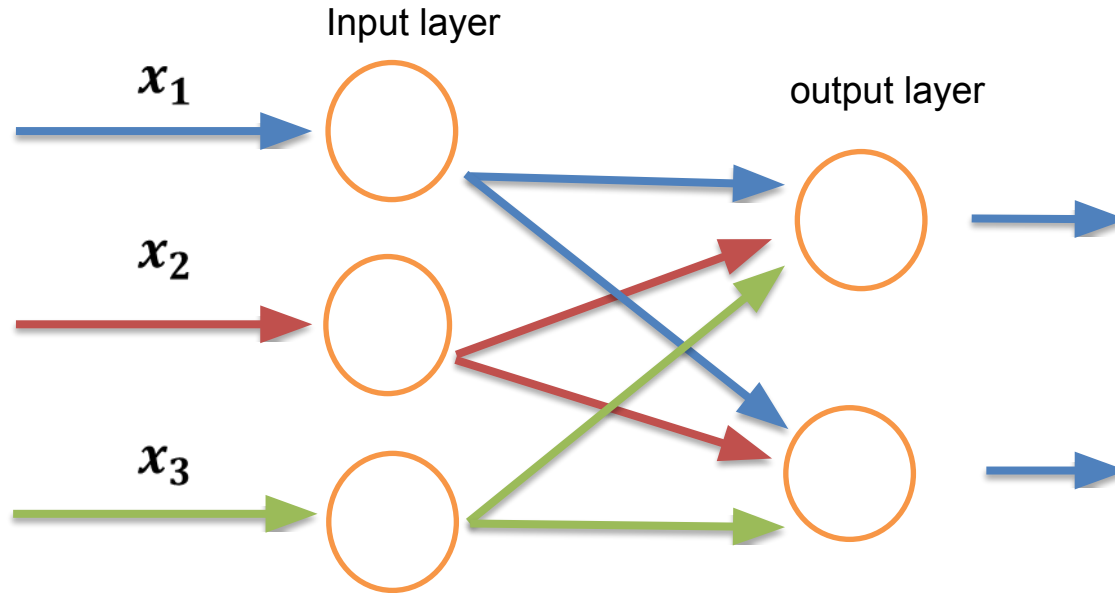


OUTPUT LAYER



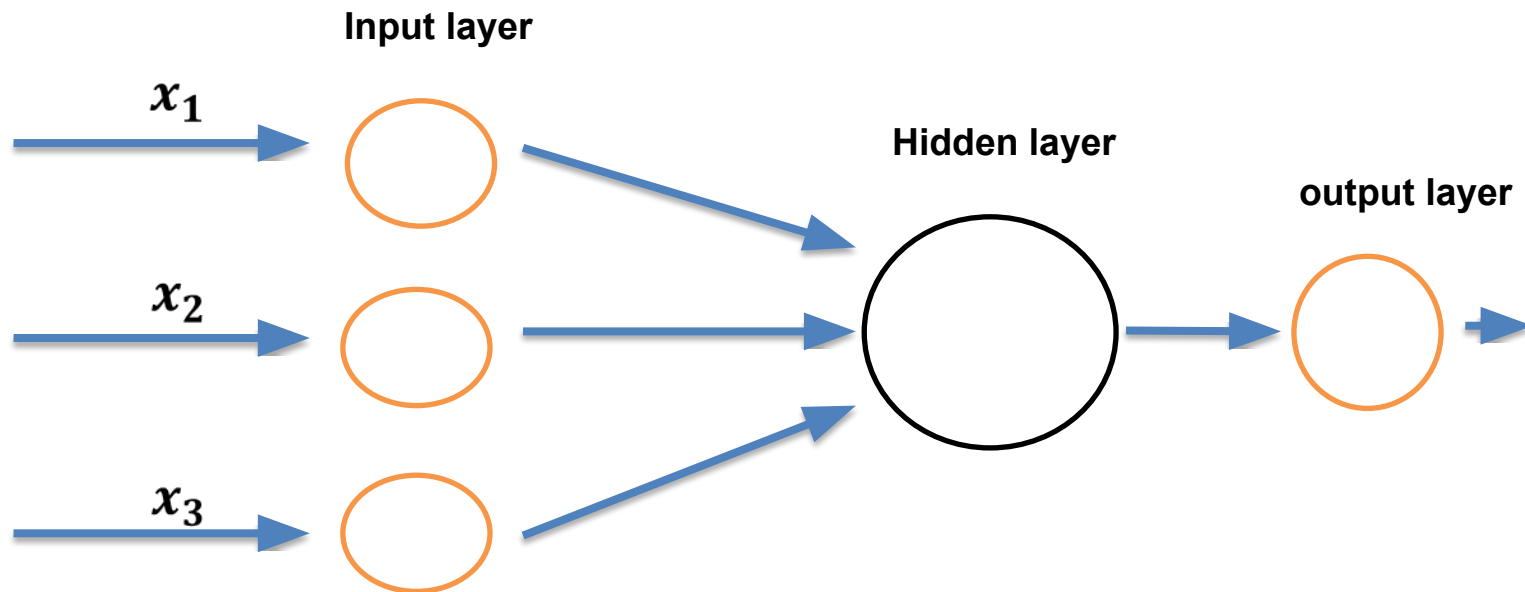
SINGLE LAYER PERCEPTRON

A neural network with single input and output layer is called single layer perceptron.



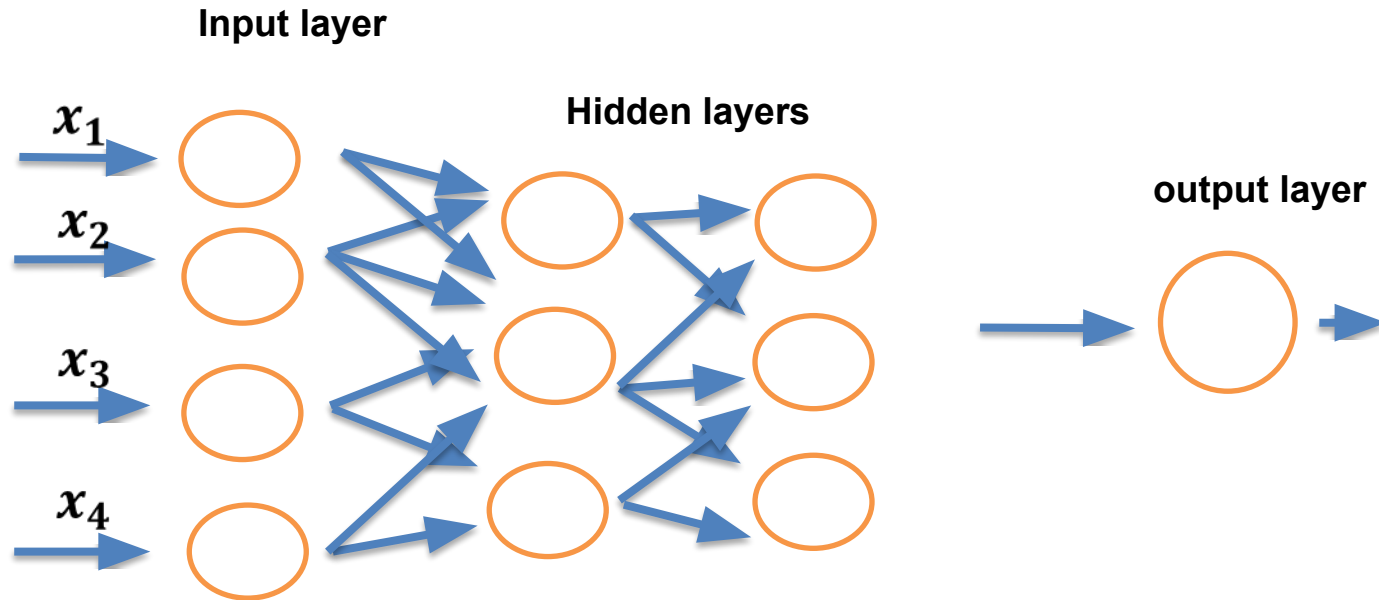
MULTI LAYER PERCEPTRON

A neural network with input layer ,output layer and one or more hidden layers is called Multi layer perceptron.



MULTI LAYER PERCEPTRON

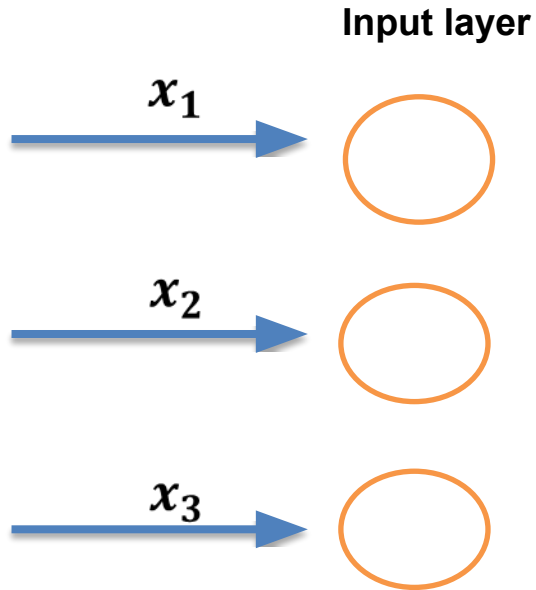
A neural network with input layer ,output layer and one or more hidden layers is called Multi layer perceptron.



HOW ANN WORKS?

STEP 1:

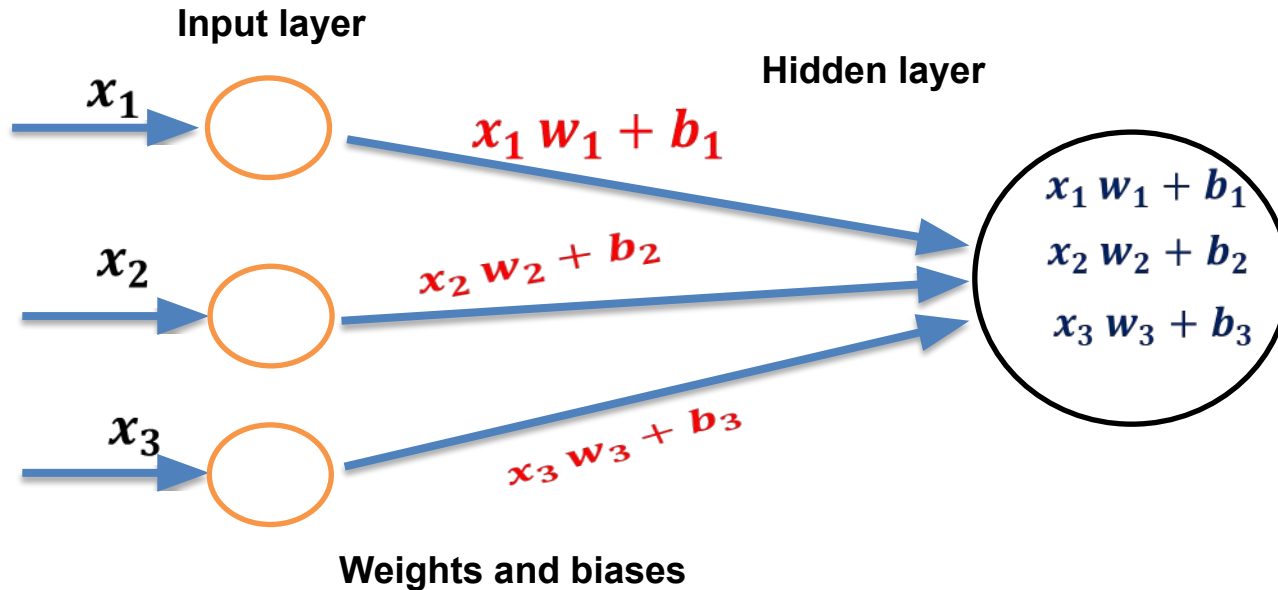
Feed data into the input layer.



HOW ANN WORKS?

STEP 2:

Data from input layer is now passed on to the hidden layer along with weights and bias.



WEIGHTS AND BIAS

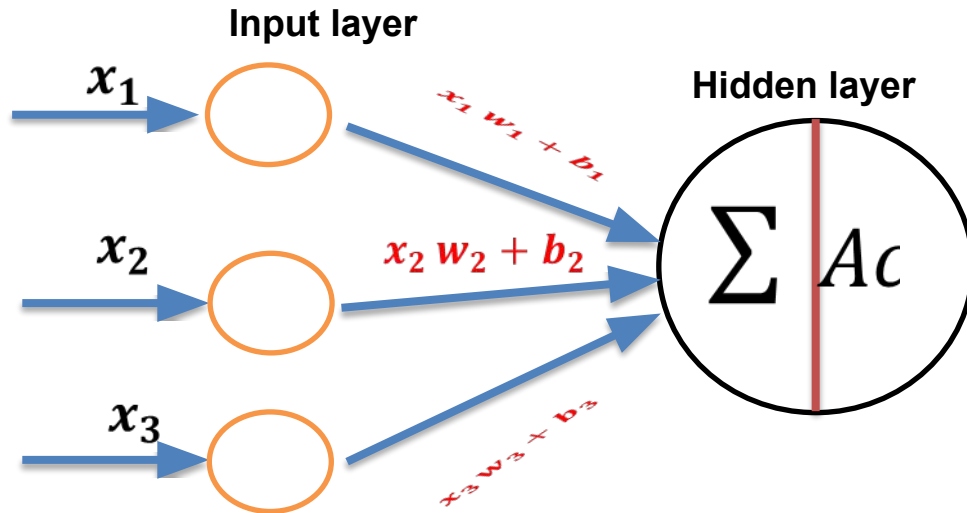
WEIGHTS

Weights are gradients of input data which are randomly initialized in the neural network.

BIAS

Some times what if weights are zero , then there might be cut down of input data from neural network which results in loss of data. To overcome that we use small constant value called as bias.

HOW ANN WORKS?



STEP 2:

There are two steps in hidden layer.

1) Summation of input data along with weights and bias.

$$f(x) = x_1 w_1 + b_1 + x_2 w_2 + b_2 + x_3 w_3 + b_3$$
$$f(x) = \sum(x_i w_i + b_i)$$

2) Activation function applied on obtained function

It is a mathematical function applied on $f(x)$ to activate the neuron.

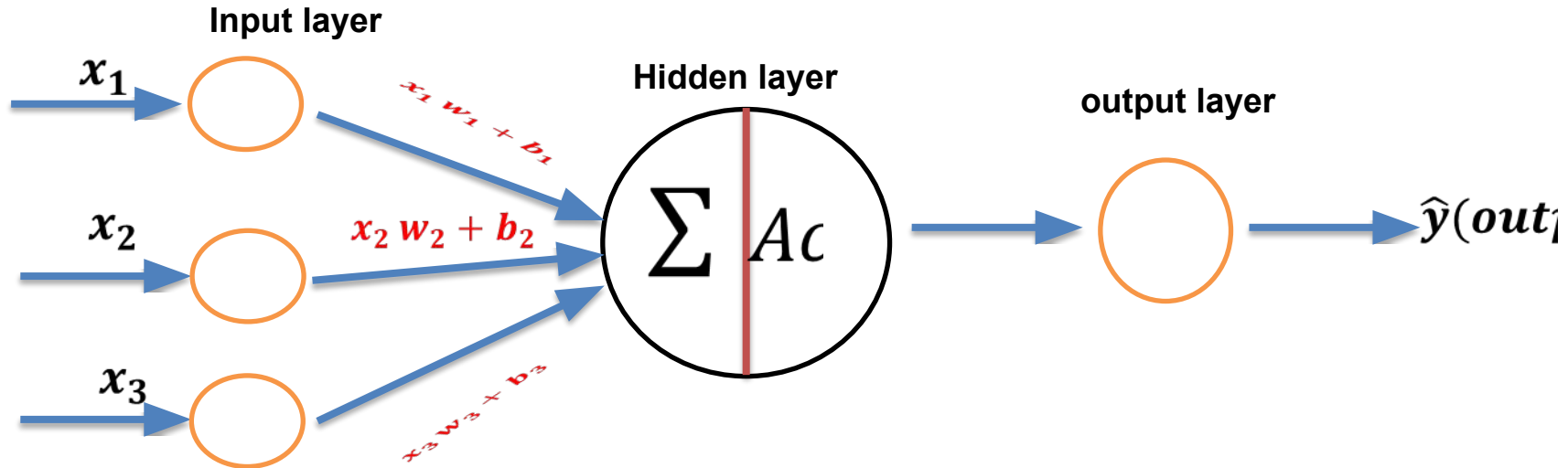
$$Act(f(x))$$

- Sigmoid
- Relu
- Softmax

HOW ANN WORKS?

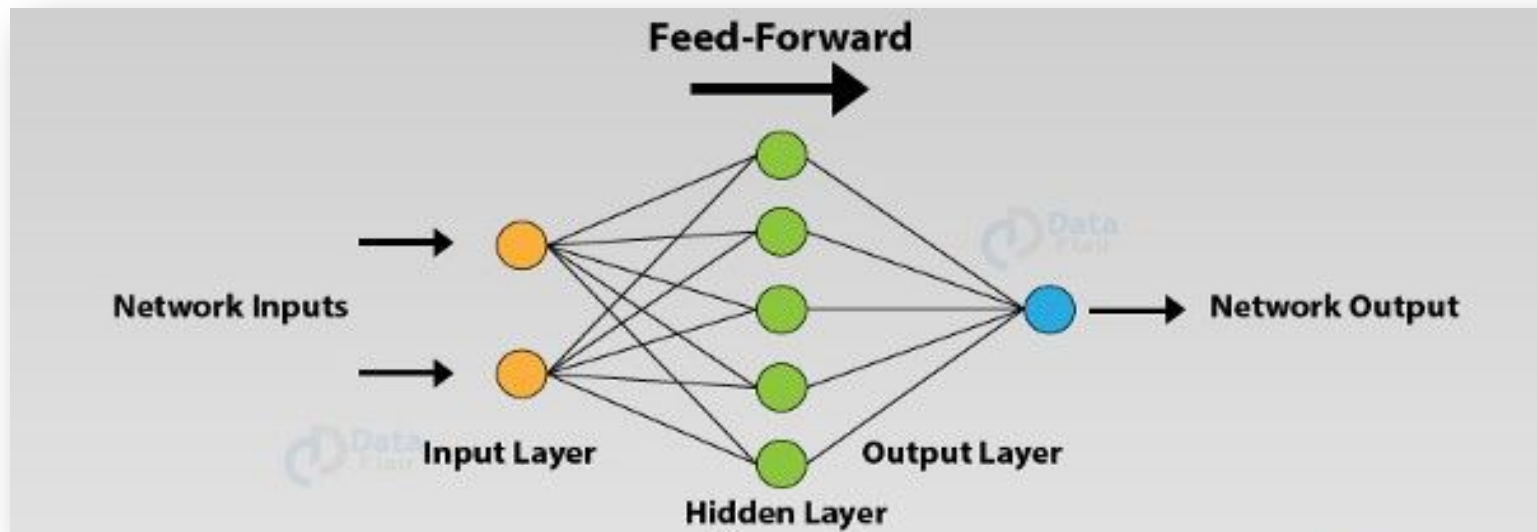
STEP 3:

Entire data is passed on to the output layer where actual prediction takes place.



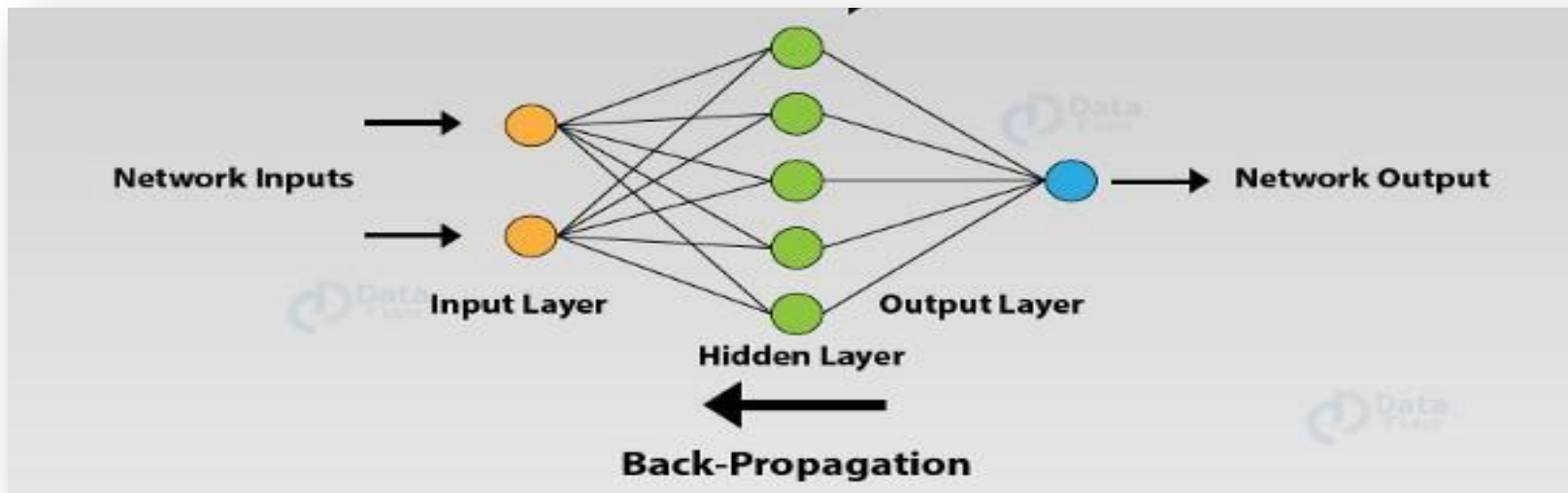
FEED FORWARD PROPAGATION

Training neural network from input layer to output layer is called as **Forward propagation**.



BACK PROPAGATION

Updating weights and bias from output layer to input layer is called as **Back propagation**.



LOSS FUNCTIONS

Why Loss function?

Loss function in neural network quantifies the difference between actual value and predicted value.

Loss functions used in Neural network are

Regression loss functions

- Mean squared error

$$MSE = \frac{1}{n} \sum (y_i - \hat{y}_i)^2$$

- Mean absolute error

$$MAE = \frac{1}{n} \sum |y_i - \hat{y}_i|$$

Classification loss functions

- Binary cross entropy /log loss

$$loss = -y_i \log P_i - (1 - y_i) \log(1 - P_i)$$

- Categorical cross entropy(softmax+CE)

$$loss = -\sum y_i \log(f(P_i))$$

OPTIMIZERS

Why Optimizers?

Optimizers in neural network are used to minimize the error/ loss by modifying the attributes of neural network such as Weights and bias.

Optimizers used in Neural network are

- Gradient descent
- Stochastic gradient descent
- Adam
- Adagrad
- RMSProp

GRADIENT DESCENT

How Gradient Descent works?

Gradient descent is an optimizer which is used to minimize the error by updating the weights.

Steps:

- Randomly start with weights and works iteratively to find optimal weight till it reaches global minima where error is less.
- It uses the concept of partial derivatives to update the weights till it reaches global minima i.e where loss is minimum

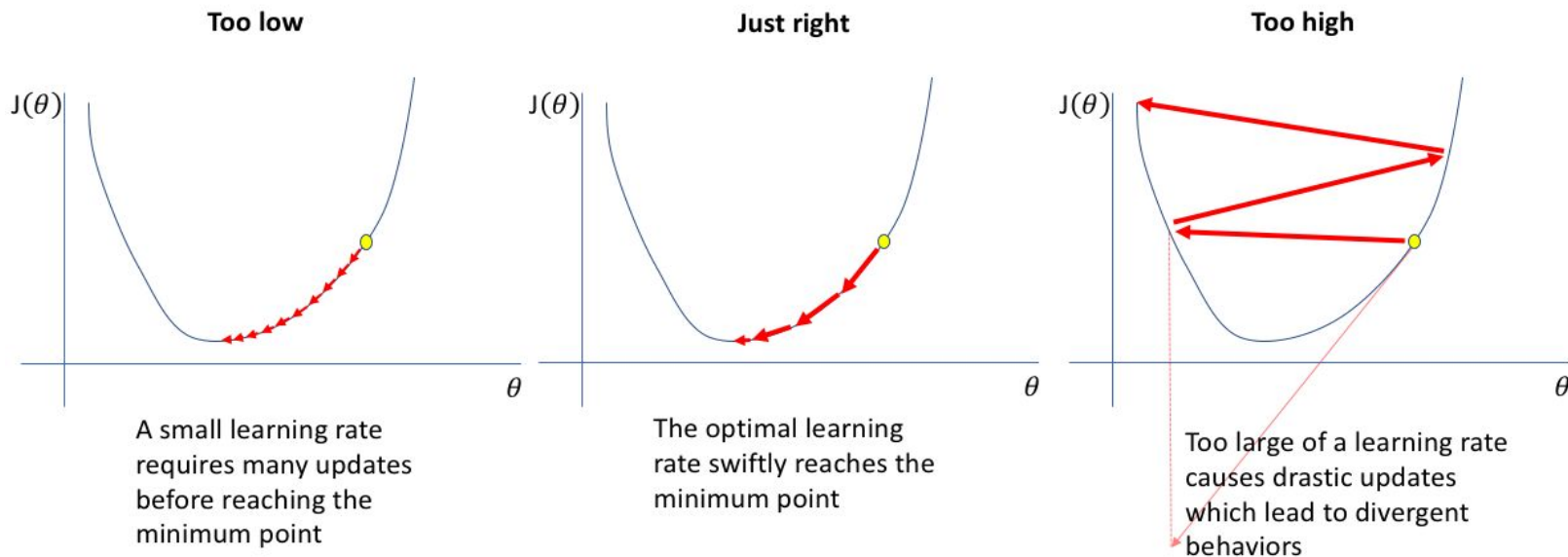
$$w_{new} = w_{old} - \alpha \frac{\partial L}{\partial w}$$

$$b_{new} = b_{old} - \alpha \frac{\partial L}{\partial b}$$



Learning Rate

Learning rate is a tuning parameter in an optimization algorithm that determines the step size at each iteration while moving toward a minimum of a loss function.



ACTIVATION FUNCTIONS

Why Activation functions?

Activation functions are the mathematical functions in neural network which help in activating the neuron.

Activation functions used in Neural network are

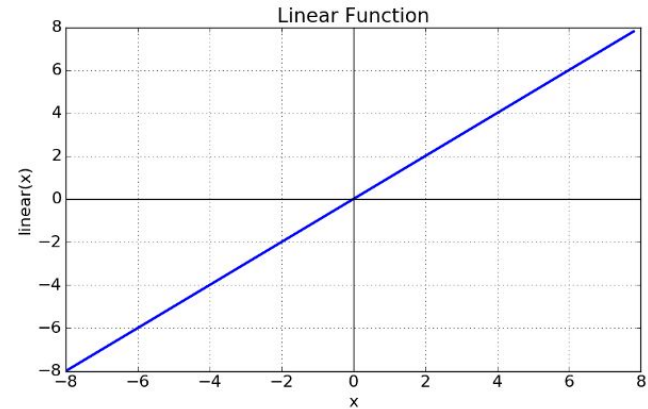
- Linear
- Sigmoid
- Softmax
- Relu
- tanh
- Leaky relu

Linear function

$$f(z) = z$$

• range = $-\infty$ to $+\infty$

- Used only for linear data.

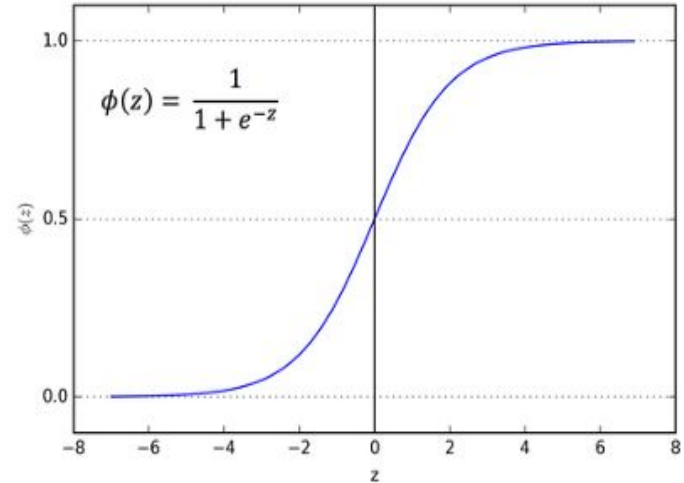


Sigmoid function

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

range=[0, 1]

- Used only for non linear data and for binary classification.
- It is used only in output layers .



Relu function

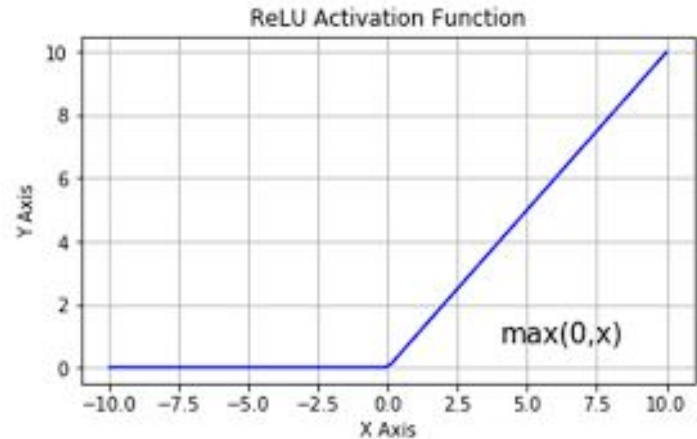
$$f(z) = \max(0, z)$$

Or

$$f(z) = \begin{cases} z, & z > 0 \\ 0, & z < 0 \end{cases}$$

$$\text{range}=[0, \infty]$$

- Most used activation function
- It is used in hidden layers .



Leaky Relu function

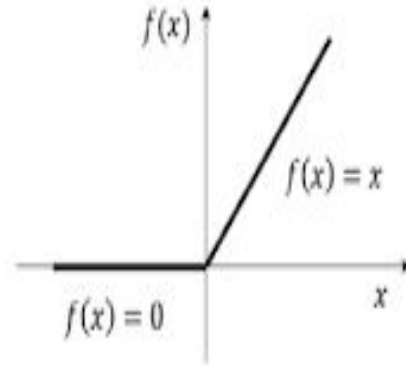
$$f(z) = \max(0.01, z)$$

Or

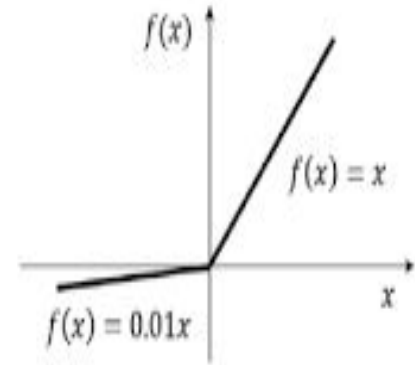
$$f(z) = \begin{cases} z, & z > 0 \\ 0.01z, & z < 0 \end{cases}$$

range= $[-\infty, \infty]$

- It is used in hidden layers .



ReLU activation function



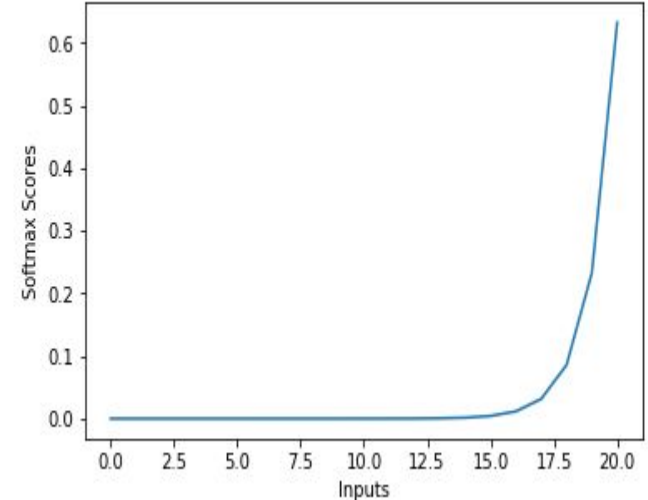
LeakyReLU activation function

Softmax function

range=[0, 1]

- Used for multiclassification problems.
- It is used only in output layers .

$$\sigma(\vec{z})_i = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}$$

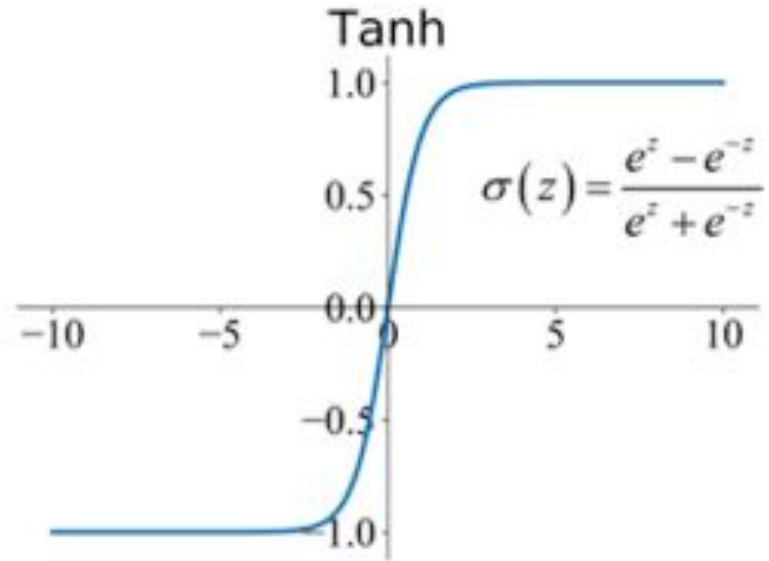


Tanh function

$$f(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$

range= $[-1, 1]$

- It is used only in hidden layers .



Hyper parameters

- **Hidden layer size**
- **Learning rate**
- **Max_iteration**

PROS

- Neural networks are flexible and can be used for both regression and classification problems.
- Neural networks are good to model with nonlinear data with large number of inputs; for example, images. It is reliable in an approach of tasks involving many features. It works by splitting the problem of classification into a layered network of simpler elements.
- Once trained, the predictions are pretty fast.
- Neural networks can be trained with any number of inputs and layers.
- Neural networks work best with more data points.

CONS

- Neural networks are black boxes, meaning we cannot know how much each independent variable is influencing the dependent variables.
- It is computationally very expensive and time consuming to train with traditional CPUs.
- Neural networks depend a lot on training data. This leads to the problem of over-fitting and generalization. The model relies more on the training data and may be tuned to the data.

APPLICATIONS

- Speech Recognition
- Handwritten character recognition
- Signature classification
- Medical domain to detect cancer cell and to analyze MRI etc.